

The Three-Dimensional Navier–Stokes Regularity Problem

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Abstract

Let u be the maximal classical solution of the three-dimensional incompressible Navier–Stokes equations on $\mathbb{R}^3$, generated by a divergence-free Schwartz datum u_0 at viscosity $\nu > 0$. The regularity problem asks whether its maximal time T_* can be finite. The threat is visible in a stretching vortex: it can narrow while kinetic energy remains finite. Scaling leaves the critical $\dot{H}^{1/2}$ height unchanged, and viscosity assigns a width r the shorter clock r^2/ν , so shrinking clocks can fit before a finite time. Yet the fluid must cross every change. Its temporal responses come from the same equation, whose viscosity couples that Tower to higher spatial derivatives.

On each strict subinterval, viscosity genuinely gains one spatial derivative in every finite response. At a proposed last time, that estimate must instead survive every approaching cutoff. The lowest payment is

$$D_* := \int_0^{T_*} \left\| \Lambda^{3/2} u(t) \right\|_2^2 dt < \infty.$$

This is equivalent to $u \in L^2(0, T_*; H^2)$, $u_t \in L^2(0, T_*; L^2)$, and to a bounded upper prefix of the signed critical production. Each form gives $u \in C([0, T_*]; H^1) \subset L^\infty(0, T_*; L^3)$; the endpoint L^3 criterion then excludes a finite T_* .

Can kinetic energy make that payment? With $Z = \|\Lambda u\|_2^2$ and $Y = \|\Lambda^2 u\|_2^2$, the direct enstrophy route stops at

$$Z' + \nu Y \leq C \nu^{-3} Z^3,$$

while energy controls only $\int Z$. Viscosity absorbs small critical data, and energy carries any history surviving beyond a datum-dependent horizon into that regime. An earlier large-data singularity remains open. Energy pays every fixed low-frequency range, so any failure of D_* is ultraviolet.

The same endpoint-uniform payment at every finite order produces one H^∞ terminal state and a same-history restart, returning conditionally to the vortex and local singularity tests. The manuscript does not derive $D_* < \infty$ from arbitrary (u_0, ν) . That ultraviolet estimate remains the missing step between the datum and global regularity.

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1 The Regularity Problem

The question here is whether the three-dimensional Navier–Stokes equation can take perfectly smooth initial data and produce a first singular time.

We start by fixing the data, the equation, and the exact global conclusion. The initial field is not allowed to carry roughness into the problem, and a solution on every shorter interval is not yet a solution through its alleged last time. Once those two points are explicit, the regularity problem becomes concrete: can the same velocity reach T_* with every derivative still finite, or can the equation force one of those derivatives to escape while kinetic energy remains bounded?

1.1 The equation and the target

Fefferman’s formulation of the whole-space problem begins with an initial velocity that leaves no roughness to blame later (Fefferman 2000). It is smooth, divergence-free, and every one of its spatial derivatives dies faster than any polynomial as $|x|$ moves out through the fluid. Write this class as

$$\mathcal{S}_\sigma := \left\{ a \in C^\infty(\mathbb{R}^3; \mathbb{R}^3) : \begin{array}{l} \nabla \cdot a = 0, \\ \forall \alpha \in \mathbb{N}_0^3 \forall K \in \mathbb{N}_0 \exists C_{\alpha,K} < \infty \forall x \in \mathbb{R}^3, \\ |\partial_x^\alpha a(x)| \leq C_{\alpha,K} (1 + |x|)^{-K} \end{array} \right\}. \quad (1.1)$$

Read the condition in the order it is written. Choose any spatial derivative α . Choose any algebraic decay rate K , however large. The datum still has a finite constant $C_{\alpha,K}$ that controls that derivative everywhere in space. The constant changes when either choice changes; it does not change with x . This is the divergence-free Schwartz class.

For a finite spatial height m , let

$$H_\sigma^m(\mathbb{R}^3) := \{v \in H^m(\mathbb{R}^3; \mathbb{R}^3) : \nabla \cdot v = 0\}.$$

The decay in (1.1) places the same datum in every one of these spaces. To see it, fix $K > 3/2$. For $|\alpha| \leq m$,

$$\|\partial_x^\alpha a\|_2^2 \leq C_{\alpha,K}^2 \int_{\mathbb{R}^3} (1 + |x|)^{-2K} dx < \infty.$$

Only finitely many derivatives occur below a chosen finite m , and hence

$$\mathcal{S}_\sigma \subset H_\sigma^m(\mathbb{R}^3) \quad \text{for every finite } m. \quad (1.2)$$

At time zero, then, every finite Sobolev height is already present. The regularity problem asks whether the evolution can destroy that complete smooth structure in finite time.

Fix one $u_0 \in \mathcal{S}_\sigma$ and one viscosity $\nu > 0$. The velocity u and the pressure p obey one coupled law:

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \quad \nabla \cdot u = 0, \quad u(\cdot, 0) = u_0. \quad (1.3)$$

The four parts should be kept together. The time derivative records the response of the field. Transport lets the velocity carry itself through the velocity it has already created. Incompressibility forces pressure to answer for that motion across the surrounding field. Viscosity acts on the spatial curvature of the same velocity. There is one fluid here, and every later derivative row is another reading of this law.

Local theory starts that fluid smoothly. Uniqueness then joins the local pieces into one maximal classical history on an interval $[0, T_*)$, where $T_* \in (0, \infty]$. The Millennium Prize problem is the claim that this interval never has a finite right edge. In the notation above, its target is

$$T_* = \infty, \quad u, p \in C^\infty(\mathbb{R}^3 \times [0, \infty)), \quad \sup_{t \geq 0} \int_{\mathbb{R}^3} |u(x, t)|^2 dx < \infty. \quad (1.4)$$

Smoothness includes the last slice of every finite interval. A field can be smooth on every shorter interval $[0, T]$, $T < T_*$, while one of its derivatives grows without a finite trace as $t \uparrow T_*$. Interior smoothness and arrival at the endpoint are different statements. The proof has to carry the derivatives to that face or show that the face cannot exist.

2 Critical Height and the Derivative Tower

The question now is what the same fluid would have to do if T_* were finite. Its kinetic energy can remain bounded, so the singular motion has to appear in the way the velocity changes across space.

Take the cleanest case: a perfectly stretching vortex, with no wobble or loss of alignment to interrupt the amplification. Keep that one vortex in view as its cross-section narrows and the viscous clock shortens until infinitely many smaller clocks appear capable of fitting inside a finite time. But naming a new scale does not move the fluid into it. The velocity has to cross the change. Its first temporal response measures that motion; if it loses control, the next asks what carried the loss. Repeating the question creates the derivative Tower: the proposed singularity must account for every response in the same fluid.

2.1 The critical-height problem inside a stretching vortex

Take a region of fluid carrying coherent rotation and place it in an extensional strain. Write

$$\omega := \nabla \times u, \quad S := \frac{1}{2}(\nabla u + (\nabla u)^\top).$$

The curl of the momentum equation gives

$$\partial_t \omega + (u \cdot \nabla) \omega = S \omega + \nu \Delta \omega. \quad (2.1.1)$$

The term $S\omega$ is the exact action of strain on vorticity. When ω points along an extensional direction of S ,

$$\omega \cdot S\omega > 0,$$

and the rotation intensifies.

The vortex does not receive that amplification without changing its shape. Incompressibility says

$$\text{tr } S = \nabla \cdot u = 0.$$

Extension along the vortex axis must be accompanied by contraction in the transverse directions. The structure becomes longer, its cross-section becomes narrower, and the velocity variation carried across that cross-section is pressed into less space. If ΔU is the variation across a width ℓ , then

$$|\nabla u| \sim \frac{\Delta U}{\ell}.$$

The regularity problem is already visible in this ratio. The amount of velocity may stay finite while the distance over which it changes collapses.

Viscosity cannot be discussed apart from that narrowing. Pair (2.1.1) with ω . The pointwise form is

$$\frac{1}{2}(\partial_t + u \cdot \nabla)|\omega|^2 = \omega \cdot S\omega + \frac{\nu}{2}\Delta|\omega|^2 - \nu|\nabla\omega|^2.$$

After integration over $\mathbb{R}^3$, transport and the Laplacian flux disappear, and the competition becomes

$$\frac{1}{2} \frac{d}{dt} \|\omega(t)\|_2^2 + \nu \|\nabla\omega(t)\|_2^2 = \int_{\mathbb{R}^3} \omega \cdot S\omega \, dx. \quad (2.1.2)$$

Stretching and diffusion are acting on the same rotating structure. As its width falls, both terms strengthen. The balance has shown us the mechanism, but it has not told us whether viscosity gains on the stretching.

Now shrink the picture without changing the law that governs it. For $\lambda > 0$, define

$$u_\lambda(x, t) := \lambda u(\lambda x, \lambda^2 t), \quad p_\lambda(x, t) := \lambda^2 p(\lambda x, \lambda^2 t). \quad (2.1.3)$$

Every term in the momentum equation receives the factor λ^3 :

$$\begin{aligned} \partial_t u_\lambda &= \lambda^3 (\partial_t u)(\lambda x, \lambda^2 t), & (u_\lambda \cdot \nabla) u_\lambda &= \lambda^3 ((u \cdot \nabla) u)(\lambda x, \lambda^2 t), \\ \nabla p_\lambda &= \lambda^3 (\nabla p)(\lambda x, \lambda^2 t), & \Delta u_\lambda &= \lambda^3 (\Delta u)(\lambda x, \lambda^2 t). \end{aligned}$$

Nothing in the smaller copy has handed viscosity an extra power of scale. Transport, pressure, and diffusion remain locked in the original proportion. The scale change has instead shortened the clock by λ^{-2} .

To see what the shrinking leaves unchanged, let $\Lambda = (-\Delta)^{1/2}$. A change of variables gives

$$\|u_\lambda(t)\|_{\dot{H}^s}^2 = \lambda^{2s-1} \|u(\lambda^2 t)\|_{\dot{H}^s}^2. \quad (2.1.4)$$

The exponent vanishes at one height:

$$2s - 1 = 0, \quad s = \frac{1}{2}.$$

We call

$$H(t) := \frac{1}{2} \|u(t)\|_{\dot{H}^{1/2}}^2 = \frac{1}{2} \|\Lambda^{1/2} u(t)\|_2^2 \quad (2.1.5)$$

the critical height. It is the spatial reading that a lawful concentration cannot make smaller by shrinking itself. Energy sits below it and loses one power of λ ; the first derivative sits above it and gains one:

$$\|u_\lambda(t)\|_2^2 = \lambda^{-1} \|u(\lambda^2 t)\|_2^2, \quad H_\lambda(t) = H(\lambda^2 t), \quad \|\nabla u_\lambda(t)\|_2^2 = \lambda \|\nabla u(\lambda^2 t)\|_2^2.$$

This is why finite energy can remain calm while a derivative sharpens. The critical height occupies the point between those two readings where the concentration is still fully visible.

Now return to the kinetic-energy identity:

$$\frac{1}{2} \|u(t)\|_2^2 + \nu \int_0^t \|\nabla u(s)\|_2^2 ds = \frac{1}{2} \|u_0\|_2^2. \quad (2.1.6)$$

The first term measures velocity over volume, one full scaling level below H . It does not retain the width across which that velocity changes. A narrowing region can lose kinetic energy while its gradients rise and its critical height remains visible.

One rescaled profile makes the separation concrete. For a nonzero divergence-free $\phi \in \mathcal{S}_\sigma$, set

$$u_\ell(x) := \ell^{-1} \phi(x/\ell).$$

Then

$$\begin{aligned} \|u_\ell\|_\infty &\sim \ell^{-1}, \quad \|\nabla u_\ell\|_\infty \sim \ell^{-2}, \\ \|u_\ell\|_2^2 &\sim \ell, \quad \|u_\ell\|_{\dot{H}^{1/2}}^2 = \|\phi\|_{\dot{H}^{1/2}}^2. \end{aligned}$$

The amplitude and gradient rise, the energy falls, and the critical height does not move. This family is not a Navier–Stokes trajectory. It shows the shape of the loophole left open by energy and respected by scaling.

The critical height tells us what a narrowing region can preserve while its energy falls. The linear heat flow supplies the other part of the picture: it assigns an exact time scale to a width. If $v_t = \nu \Delta v$, then

$$\hat{v}(\xi, t + \delta t) = e^{-\nu |\xi|^2 \delta t} \hat{v}(\xi, t).$$

A structure of width r lives near $|\xi| \simeq r^{-1}$. Diffusion changes its amplitude by order one on the time

$$\tau_\nu(r) \asymp \frac{r^2}{\nu}. \quad (2.1.7)$$

This is not yet an estimate on the nonlinear fluid. It is the amount of time viscosity assigns to a scale once the evolution has actually formed that scale.

Take geometrically shrinking widths

$$r_n := 2^{-n}r_0, \quad \tau_n := \frac{r_n^2}{\nu}.$$

Their heat clocks satisfy

$$\tau_n = \frac{r_0^2}{\nu} 4^{-n}, \quad \sum_{n=0}^{\infty} \tau_n = \frac{4r_0^2}{3\nu} < \infty. \quad (2.1.8)$$

This is the Zeno opening: the clocks attached to infinitely many smaller scales fit inside a finite amount of elapsed time.

The arithmetic does not create a singularity. The fluid would have to form r_0 , then r_1 , then r_2 , continuing without losing the stretching alignment while viscosity is already acting on every scale that has appeared. Each new event would begin from the velocity produced by the previous one. The widths shrink, the clocks shorten, and the critical stretching–diffusion balance has to be reproduced again.

That is what turns the spatial picture into a question about response. If the velocity is to cross sharper states during shorter intervals, what must its acceleration do? If the acceleration is forced upward, what must lead it? The singularity has brought us to the temporal derivative Tower.

2.2 Velocity, acceleration, jerk, and the Tower

The Zeno picture compresses space and time together. A smaller width is not reached by naming a later scale; the velocity has to move through the change during the shorter interval available. Let X be a spatial Banach space in which the classical responses are defined on every strict subinterval. For each $n \geq 0$,

$$\|\partial_t^n u(t_1) - \partial_t^n u(t_0)\|_X \leq \int_{t_0}^{t_1} \|\partial_t^{n+1} u(s)\|_X ds. \quad (2.2.1)$$

If $\partial_t^{n+1} u$ stayed uniformly bounded before a finite T_* , this inequality would keep $\partial_t^n u$ uniformly bounded as well. Reading that implication upward, any escape of u in X forces every finite temporal response to lose uniform control in the same space:

$$\sup_{0 < t < T_*} \|u(t)\|_X = \infty \implies \sup_{0 < t < T_*} \|\partial_t^n u(t)\|_X = \infty \quad \text{for every finite } n. \quad (2.2.2)$$

This is a necessary consequence of a proposed escape, not a proof that the escape occurs. It tells us why the temporal responses have to be examined together:

$$u, \quad \partial_t u, \quad \partial_t^2 u, \quad \partial_t^3 u, \quad \dots$$

Write

$$V_n := \partial_t^n u, \quad Q_n := \partial_t^n p, \quad n \in \mathbb{N}_0. \quad (2.2.3)$$

These are the velocity, temporal acceleration, temporal jerk, and higher Eulerian responses of the fluid. The material acceleration is the full combination

$$D_t u = \partial_t u + (u \cdot \nabla) u;$$

the Tower follows $\partial_t^n u$ at fixed spatial position and keeps the transport term inside the recurrence below. The Tower is not an extra model placed beside Navier–Stokes. The original equation generates every rung.

At the base,

$$V_1 = \nu \Delta V_0 - (V_0 \cdot \nabla) V_0 - \nabla Q_0.$$

Differentiate once:

$$V_2 = \nu \Delta V_1 - (V_1 \cdot \nabla) V_0 - (V_0 \cdot \nabla) V_1 - \nabla Q_1.$$

Differentiate again, and the two ways of placing one derivative on each transport factor produce the middle coefficient:

$$V_3 = \nu \Delta V_2 - (V_2 \cdot \nabla) V_0 - 2(V_1 \cdot \nabla) V_1 - (V_0 \cdot \nabla) V_2 - \nabla Q_2.$$

The general row is the binomial expansion of the transport term:

$$V_{n+1} + \sum_{a=0}^n \binom{n}{a} (V_a \cdot \nabla) V_{n-a} + \nabla Q_n = \nu \Delta V_n, \quad \nabla \cdot V_n = 0. \quad (2.2.4)$$

Pressure does not create a second Tower. Taking the divergence of the base equation and using the decaying whole-space normalization gives

$$-\Delta p = \partial_i \partial_j (u_i u_j), \quad p = R_i R_j (u_i u_j), \quad (2.2.5)$$

where $R_j = \partial_j (-\Delta)^{-1/2}$. After n time derivatives,

$$Q_n = R_i R_j \sum_{a=0}^n \binom{n}{a} (V_a)_i (V_{n-a})_j. \quad (2.2.6)$$

The velocity rows determine Q_n ; its gradient returns to (2.2.4) as part of the next response. Every rung retains transport, pressure, viscosity, and incompressibility.

The Tower is now fixed by the original equation: velocity generates acceleration, acceleration generates jerk, and every later response retains transport, pressure, incompressibility, and viscosity. One question has been left inside the temporal law. Each rung contains $\nu \Delta V_n$. As the responses climb in time, what does that Laplacian force us to carry in space?

3 The Intersection Inside the Tower

The Tower was built because a finite-time singularity would have to force the same velocity through faster and faster responses. Yet the equation that generates the next response contains $\nu\Delta V_n$: two spatial derivatives of the response already in hand. The attempted temporal escape cannot be read only in time. We have to ask whether climbing the Tower spends spatial regularity or whether a higher spatial energy appears with viscosity in the same calculation.

Put the two directions beside each other. Time runs down through $V_0, V_1, V_2, \dots$; space runs across the Sobolev heights of each response. A dot in the table names a mixed derivative of the same velocity. It does not yet say that the derivative stays bounded near a final time. We begin at the critical height and let the equation show exactly what viscosity exposes.

3.1 The temporal–spatial table

The vertical direction is already generated by time differentiation. The horizontal direction is the one viscosity must now justify.

Table 3.1.1: The two directions inside the Tower. Each bullet names a mixed derivative of the same velocity; none is yet an estimate.

	H_x^0	H_x^1	H_x^2	H_x^3	$\dots$
$V_0 = u$	•	•	•	•	$\dots$
$V_1 = u_t$	•	•	•	•	$\dots$
$V_2 = u_{tt}$	•	•	•	•	$\dots$
$V_3 = u_{ttt}$	•	•	•	•	$\dots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$

3.2 The spatial row inside the first response

The table has placed temporal response against spatial height. Start with the base response $V_0 = u$, where the two directions first became visible at critical height. For $s \geq 0$, set

$$E_s(t) := \frac{1}{2} \|\Lambda^s u(t)\|_2^2$$

and retain the complete transport–pressure pairing

$$\mathcal{P}_s(t) := -\operatorname{Re} \left\langle \Lambda^s u, \Lambda^s \left((u \cdot \nabla) u + \nabla p \right) \right\rangle_{L^2}.$$

The spatial energy row is

$$E'_s = \mathcal{P}_s - 2\nu E_{s+1}. \tag{3.2.1}$$

At $s = \frac{1}{2}$,

$$H' = \mathcal{P}_{1/2} - 2\nu E_{3/2}. \quad (3.2.2)$$

The next two differentiated balances make the triangular structure explicit:

$$H'' = \partial_t \mathcal{P}_{1/2} - 2\nu \mathcal{P}_{3/2} + (2\nu)^2 E_{5/2}, \quad (3.2.3)$$

$$H''' = \partial_t^2 \mathcal{P}_{1/2} - 2\nu \partial_t \mathcal{P}_{3/2} + (2\nu)^2 \mathcal{P}_{5/2} - (2\nu)^3 E_{7/2}. \quad (3.2.4)$$

Every lower transfer survives while one higher spatial energy appears.

The complete identity keeps every lower transfer while exposing one new spatial height.

Proposition 3.2.1 (Critical-height spatial ascent). *For every integer $n \geq 1$,*

$$H^{(n)} = (-2\nu)^n E_{n+1/2} + \sum_{j=0}^{n-1} (-2\nu)^j \partial_t^{n-1-j} \mathcal{P}_{j+1/2}, \quad n \geq 1. \quad (3.2.5)$$

Proof. The critical balance (3.2.2) is the case $n = 1$. Assume the identity at order n , differentiate it, and apply (3.2.1) at height $n + \frac{1}{2}$. The new transfer extends the sum by one term, while the viscous contribution becomes $(-2\nu)^{n+1} E_{n+3/2}$. This is the identity at order $n + 1$. $\square$

Rearranging for the highest energy gives

$$E_{n+1/2} = \frac{H^{(n)} - \sum_{j=0}^{n-1} (-2\nu)^j \partial_t^{n-1-j} \mathcal{P}_{j+1/2}}{(-2\nu)^n}. \quad (3.2.6)$$

The division is lawful because $\nu > 0$. It does not erase the transfer sum. This identity names the exact collection that has to be controlled before the exposed spatial energy is controlled.

The height derivative remains a contraction of the velocity responses:

$$H^{(n)} = \frac{1}{2} \sum_{a=0}^n \binom{n}{a} \operatorname{Re} \langle \Lambda^{1/2} V_a, \Lambda^{1/2} V_{n-a} \rangle_{L^2}. \quad (3.2.7)$$

The three readings at order n are distinct. V_n is the Eulerian response, $H^{(n)}$ is its quadratic critical contraction, and $E_{n+1/2}$ is the highest spatial energy exposed after the full transfer chain has been retained. The identity reveals that energy; it does not bound the transfer chain on its right. This is the exact point at which the hoped-for reversal becomes conditional. Viscosity has exposed a higher spatial energy; writing the identity has not controlled that energy.

Does the same obstruction remain after the base response changes? Shift the calculation from u to a fixed V_r . The Laplacian still acts linearly on that response, while every split of the transport derivative remains in the binomial sum. The same calculation now answers the question for all finite r .

3.3 The same spatial gain inside every response

Fix $r \in \mathbb{N}_0$. The scalar energy of the response V_r at spatial height s is

$$E_{r,s} := \frac{1}{2} \|\Lambda^s V_r\|_2^2.$$

The transfer paired with that response is

$$\mathcal{P}_{r,s} := -\operatorname{Re} \left\langle \Lambda^s V_r, \Lambda^s \left[\sum_{a=0}^r \binom{r}{a} (V_a \cdot \nabla) V_{r-a} + \nabla Q_r \right] \right\rangle_{L^2}.$$

Pairing the r -th differentiated momentum equation at this height produces

$$E'_{r,s} = \mathcal{P}_{r,s} - 2\nu E_{r,s+1}. \quad (3.3.1)$$

Beginning at $s = \frac{1}{2}$ and repeating the same induction gives the differentiated balance

$$\partial_t^n E_{r,1/2} = (-2\nu)^n E_{r,n+1/2} + \sum_{j=0}^{n-1} (-2\nu)^j \partial_t^{n-1-j} \mathcal{P}_{r,j+1/2}. \quad (3.3.2)$$

The same pattern now repeats inside every response. Moving down changes the temporal response; moving across exposes a higher spatial energy; and the nonlinear transfer remains beside it. Exposure is not yet payment. To see where the spatial gain is actually made, apply the Leray projection to the differentiated equation. It annihilates ∇Q_r and commutes with ∂_t and Δ , so

$$\partial_t V_r - \nu \Delta V_r = -F_r, \quad F_r := \mathbb{P} \sum_{a=0}^r \binom{r}{a} (V_a \cdot \nabla) V_{r-a}.$$

For an integer $m \geq 2$, the Sobolev multiplication used in Proposition A.1.1 gives

$$\|F_r\|_{H^{m-1}} \leq C_{m,r} \sum_{a=0}^r \|V_a\|_{H^m} \|V_{r-a}\|_{H^m}. \quad (3.3.3)$$

Pairing the equation one spatial step across this forcing, as in the linear estimate inside Proposition A.1.1, yields

$$\frac{d}{dt} \|V_r\|_{H^m}^2 + \nu \|V_r\|_{H^{m+1}}^2 \leq C_{\nu,m} \left(\|V_r\|_{H^m}^2 + \|F_r\|_{H^{m-1}}^2 \right). \quad (3.3.4)$$

This is the local spatial payment. On a fixed strict interval $I = (0, T)$, $T < T_*$, persistence of the same classical history in Proposition A.1.1 gives, for the finite indices now in use,

$$\max_{0 \leq a \leq r} \|V_a\|_{L^\infty(0,T;H^m)} < \infty. \quad (3.3.5)$$

Thus (3.3.3) makes the right-hand side of (3.3.4) integrable. This is the unexpected turn in the table: climbing to the next response does not simply spend the spatial regularity already present. Viscosity places V_r one degree higher, while the equation places its next response one degree lower:

$$V_r \in L^2(I; H^{m+1}), \quad V_{r+1} = \partial_t V_r \in L^2(I; H^{m-1}). \quad (3.3.6)$$

Now ask for any finite q time derivatives and k spatial derivatives. Repeat the payment through finitely many responses and choose

$$R > q + \frac{1}{2}, \quad M > k + \frac{3}{2}.$$

The vector-valued time and three-dimensional spatial embeddings (Cerdà 2010, Theorem 7.28) give

$$H^R(I; H^M(\mathbb{R}^3)) \hookrightarrow C^q(\bar{I}; C^k(\mathbb{R}^3)).$$

Since q and k were arbitrary, the fixed strict interval carries

$$\mathcal{V}_\infty(I) := \bigcap_{r=0}^{\infty} \bigcap_{m=0}^{\infty} H^r(I; H^m(\mathbb{R}^3)) \hookrightarrow C^\infty(\bar{I} \times \mathbb{R}^3). \quad (3.3.7)$$

Every finite position in the grid is now filled, but only after the strict cutoff $T < T_*$ has been fixed. Neither (3.3.2) nor (3.3.4) makes it uniform as $T \uparrow T_*$.

The constants may depend on the selected (r, m) . Smoothness asks for each finite derivative in turn; it does not ask for one constant uniform over the entire grid. This is the regularity carried by one history before a chosen cutoff. It is not yet the compatible intersection at the terminal face.

A dot can now be finite on every strict subinterval and still diverge as the endpoint is approached. To carry a spatial derivative to one time slice, we have to place it beside the temporal response directly below it. Those adjacent pairs determine the finite rectangles.

3.4 The finite piece that reaches a time slice

Let $I_* = (0, T_*)$ denote the interval on a proposed finite maximal branch, and first stop at a strict cutoff $T < T_*$. Control of v one spatial step above H^M has no reason by itself to arrive continuously at $t = T$. Its time derivative must be controlled one spatial step below H^M as well.

Let $A = (1 - \Delta)^{1/2}$. For a smooth Hilbert-valued function v ,

$$\|v(b)\|_{H^M}^2 - \|v(a)\|_{H^M}^2 = 2 \operatorname{Re} \int_a^b \langle A^{M-1} v_t(t), A^{M+1} v(t) \rangle_{L^2} dt. \quad (3.4.1)$$

Cauchy–Schwarz places the middle trace between these two adjacent spatial controls

$$v \in L^2(0, T; H^{M+1}), \quad v_t \in L^2(0, T; H^{M-1}), \quad (3.4.2)$$

and the conclusion is

$$v \in C([0, T]; H^M); \quad (3.4.3)$$

the approximation argument and strong representative are proved in Theorem A.3.1.

At $M = 2$, one concrete adjacent pair is

$$u \in L^2(0, T; H^3), \quad u_t \in L^2(0, T; H^1) \implies u \in C([0, T]; H^2). \quad (3.4.4)$$

The velocity lies one spatial step above the desired trace and its next temporal response lies one spatial step below it. Their pairing reaches the middle.

Now apply that trace to V_n , whose derivative is V_{n+1} . To retain the responses through order N at a common height M , collect

$$\begin{aligned} \mathfrak{R}_{N,M}(u; T) := & \sum_{n=0}^N \|V_n\|_{L^2(0, T; H^{M+1})}^2 \\ & + \sum_{n=0}^N \|V_{n+1}\|_{L^2(0, T; H^{M-1})}^2. \end{aligned} \quad (3.4.5)$$

When $M = 0$, the lower edge is read in H^{-1} . The temporal depth N and spatial height M remain independent choices. The number $\mathfrak{R}_{N,M}(u; T)$ measures the size of this finite view of the history.

The $(N, M) = (1, 2)$ example contains two adjacent traces sharing V_1 :

Table 3.4.1: The $(1, 2)$ rectangle. The shared response closes the first trace and begins the second.

response	above the trace	below the trace
$V_0 = u$	$V_0 \in L^2 H^3$	$V_1 \in L^2 H^1$
$V_1 = u_t$	$V_1 \in L^2 H^3$	$V_2 \in L^2 H^1$

The norm alone does not remember which field produced these controls. Write $\mathcal{R}_{N,M}[u_0; T]$ for the velocity block together with the pressure responses reconstructed by (2.2.6). Every entry is a derivative of the same velocity or the normalized pressure forced by that velocity.

Classical smoothness on a strict subinterval makes each chosen rectangle finite:

$$\mathfrak{R}_{N,M}(u; T) < \infty \quad (T < T_*, N, M < \infty). \quad (3.4.6)$$

But the terminal question changes the order of the quantifiers:

$$\forall T < T_* : \quad \mathfrak{R}_{N,M}(u; T) < \infty \quad \not\Rightarrow \quad \sup_{0 < T < T_*} \mathfrak{R}_{N,M}(u; T) < \infty.$$

Fixing (N, M) first leaves one demand: a bound that survives every cutoff approaching T_* . This is the whole-terminal rectangle estimate.

4 The Missing Estimate and Global Smoothness

The table and its finite rectangles have reduced the endpoint to one demand: fix the rectangle first, then find a bound that survives every cutoff. The lowest rectangle is enough to expose where that payment must come from.

4.1 The estimate that must survive the endpoint

At $(N, M) = (0, 1)$, the velocity is controlled one spatial step above H^1 and its time response one step below. Together they can reach one time slice. This lowest rectangle asks for

$$u \in L^2(I_*; H^2), \quad u_t \in L^2(I_*; L^2). \quad (4.1.1)$$

Which part of the equation could produce these two controls? Set

$$\begin{aligned} Z(t) &:= \|\Lambda u(t)\|_2^2 = \|\omega(t)\|_2^2, \quad Y(t) := \|\Lambda^2 u(t)\|_2^2 = \|\nabla \omega(t)\|_2^2, \\ D(t) &:= \|\Lambda^{3/2} u(t)\|_2^2. \end{aligned} \quad (4.1.2)$$

The enstrophy balance (2.1.2) contains the stretching term that must be controlled. Since ∇u is recovered from ω by Riesz transforms, Hölder followed by the Sobolev embeddings produces

$$\begin{aligned} \left| \int_{\mathbb{R}^3} \omega \cdot S\omega \, dx \right| &\leq \|\omega\|_2 \|\omega\|_3 \|S\|_6 \\ &\leq CZ^{1/2} D^{1/2} Y^{1/2}. \end{aligned} \quad (4.1.3)$$

Young's inequality then turns the enstrophy identity into

$$Z'(t) + \nu Y(t) \leq C\nu^{-1} D(t) Z(t). \quad (4.1.4)$$

The coefficient multiplying Z has now been isolated. If

$$D_* := \int_0^{T_*} D(t) \, dt = \int_0^{T_*} \|\Lambda^{3/2} u(t)\|_2^2 \, dt < \infty, \quad (4.1.5)$$

then Gronwall gives

$$\sup_{0 < t < T_*} Z(t) + \int_0^{T_*} Y(t) \, dt < \infty. \quad (4.1.6)$$

The first term supplies $u \in L^\infty(I_*; H^1)$, and the second, together with the kinetic-energy row, supplies $u \in L^2(I_*; H^2)$. Applying the Leray projection to the momentum equation gives

$$u_t = \nu \Delta u - \mathbb{P}((u \cdot \nabla)u),$$

while

$$\|\mathbb{P}((u \cdot \nabla)u)\|_2^2 \leq C \|u\|_6^2 \|\nabla u\|_3^2 \leq CZD. \quad (4.1.7)$$

Thus $D_* < \infty$ also gives $u_t \in L^2(I_*; L^2)$, which completes (4.1.1). In the other direction, the cutoff-uniform $L^2 H^2$ row and monotone convergence give $u \in L^2(I_*; H^2)$, hence $D_* < \infty$. Using the energy identity in Proposition A.2.1 for the lower Sobolev pieces, we have reached the exact equivalence

$$D_* < \infty \iff \sup_{0 < T < T_*} \mathfrak{R}_{0,1}(u; T) < \infty. \quad (4.1.8)$$

The critical-height balance gives the same interface in its original coordinates. Because $D = 2E_{3/2}$, integrating (3.2.2) yields

$$H(t) + \nu \int_0^t D(s) ds = H(0) + \int_0^t \mathcal{P}_{1/2}(s) ds. \quad (4.1.9)$$

Since $H \geq 0$, finiteness of the signed production prefix makes D_* finite. Conversely, $D_* < \infty$ gives the first rectangle and hence a bounded H ; the same equality then makes the prefix finite. Hence

$$\sup_{0 < t < T_*} \int_0^t \mathcal{P}_{1/2}(s) ds < \infty \iff D_* < \infty \iff \sup_{0 < T < T_*} \mathfrak{R}_{0,1}(u; T) < \infty. \quad (4.1.10)$$

The signed-prefix statement is another exact reading of the first rectangle; it is not a smaller estimate that produces the rectangle for free.

There is one genuine class of data for which this interface closes. Its boundary sits inside the critical balance itself. Put

$$a(t) := \left\| \Lambda^{1/2} u(t) \right\|_2, \quad d(t) := \left\| \Lambda^{3/2} u(t) \right\|_2^2.$$

For $\mathcal{B}(u, u) := \mathbb{P}((u \cdot \nabla)u)$, the dual Sobolev embeddings give

$$\begin{aligned} \|\mathcal{B}(u, u)\|_{\dot{H}^{-1/2}} &\leq C \|u \cdot \nabla u\|_{3/2} \\ &\leq C \|u\|_3 \|\nabla u\|_3 \leq Ca(t)d(t)^{1/2}. \end{aligned} \quad (4.1.11)$$

The critical production pairs this source with one further half derivative of the velocity, so

$$|\mathcal{P}_{1/2}(t)| \leq Ca(t)d(t). \quad (4.1.12)$$

Inserted into (3.2.2), this becomes

$$\frac{1}{2} \frac{d}{dt} a(t)^2 + (\nu - Ca(t))d(t) \leq 0. \quad (4.1.13)$$

If $a(0) < \nu/C$, continuity prevents a from reaching the viscous threshold: until a first crossing, the coefficient is positive and a is nonincreasing, which rules out that crossing. Consequently

$$\frac{1}{2} a(t)^2 + (\nu - Ca(0)) \int_0^t d(s) ds \leq \frac{1}{2} a(0)^2. \quad (4.1.14)$$

This pays D_* from the datum in the small-critical-data regime. For an arbitrary Schwartz datum, $a(0)$ need not lie below the threshold. The source and the dissipation have the same critical scaling, so shrinking the picture does not improve the coefficient in (4.1.13).

Energy can still carry the same history into that regime if the classical branch survives long enough. The resulting time does not predict when a singularity occurs. It is a horizon beyond which a finite maximal time cannot sit.

Corollary 4.1.1 (Small-data capture horizon). *Let C be the constant in (4.1.13). If a nonzero datum $u_0 \in \mathcal{S}_\sigma$ generates a finite maximal time, then*

$$T_* \leq \frac{C^4 \|u_0\|_2^4}{2\nu^5}. \quad (4.1.15)$$

Proof. Fix $0 < T < T_*$. The kinetic-energy identity in Proposition A.2.1 gives

$$\int_0^T Z(t) dt \leq \frac{\|u_0\|_2^2}{2\nu},$$

so at some $t_0 \in (0, T)$,

$$Z(t_0) \leq \frac{\|u_0\|_2^2}{2\nu T}.$$

Interpolation between the energy and enstrophy heights then gives

$$a(t_0)^2 \leq \|u(t_0)\|_2 \|\Lambda u(t_0)\|_2, \quad a(t_0) \leq \frac{\|u_0\|_2}{(2\nu T)^{1/4}}.$$

If $T > C^4 \|u_0\|_2^4 / (2\nu^5)$, this places the shifted history below the threshold ν/C . The same first-crossing argument, now begun at t_0 , gives

$$\frac{1}{2}a(t)^2 + (\nu - Ca(t_0)) \int_{t_0}^t d(s) ds \leq \frac{1}{2}a(t_0)^2, \quad t_0 \leq t < T_*.$$

This pays the remaining critical dissipation. The portion before t_0 is finite by classical smoothness, so $D_* < \infty$. By (4.1.6), the same history is then bounded in H^1 , hence in L^3 , all the way to the alleged last time. That contradicts the Escauriaza–Seregin–Šverák endpoint criterion (4.1.23). No such $T < T_*$ can exist, which proves (4.1.15). $\square$

The kinetic-energy identity in Proposition A.2.1 supplies only $\int_0^{T_*} Z(t) dt < \infty$. Interpolating $D \leq Z^{1/2} Y^{1/2}$ in (4.1.3) shows the gap directly:

$$Z'(t) + \nu Y(t) \leq C\nu^{-3} Z(t)^3. \quad (4.1.16)$$

Control of $\int Z$ does not control $\int Z^3$ or $\sup Z$. The two calculations meet at the same boundary: neither the energy row nor the critical row bounds the quantities in (4.1.10) for an arbitrary pair (u_0, ν) .

Energy still tells us what a failure of that bound would have to look like. Interpolation between the energy and enstrophy heights gives

$$H(t)^2 = \frac{1}{4} \left\| \Lambda^{1/2} u(t) \right\|_2^4 \leq \frac{1}{4} \|u(t)\|_2^2 Z(t).$$

After the kinetic-energy identity is integrated once more in time,

$$\int_0^{T_*} H(t)^2 dt \leq \frac{\|u_0\|_2^4}{8\nu}. \quad (4.1.17)$$

This is not a pointwise roof. It says that large critical height can occupy only a short amount of time. The interpolation between the energy height and the critical dissipation makes the simultaneous spatial cost visible as well: for a nonzero datum,

$$\|\Lambda^{1/2}u(t)\|_2 \leq \|u(t)\|_2^{2/3} \|\Lambda^{3/2}u(t)\|_2^{1/3} \implies D(t) \geq \frac{8H(t)^3}{\|u_0\|_2^4}. \quad (4.1.18)$$

Now take a height $M \geq H(0)$ that the history crosses on its way to $2M$. Let t_1 be its first arrival at $2M$, and let $t_0 < t_1$ be its last visit to M before that arrival. On $I_M = (t_0, t_1)$, the height stays between M and $2M$. Equation (4.1.17) gives

$$|I_M| \leq \frac{\|u_0\|_2^4}{8\nu M^2}. \quad (4.1.19)$$

At the same time, $|\mathcal{P}_{1/2}| \leq C\sqrt{2H} D$. Integrating the critical balance across this first-hitting interval gives

$$M = \int_{I_M} H'(t) dt \leq 2C\sqrt{M} \int_{I_M} D(t) dt, \quad \int_{I_M} D(t) dt \geq \frac{\sqrt{M}}{2C}. \quad (4.1.20)$$

This is the rigorous Zeno remnant. Higher records are forced into shorter intervals, while each completed doubling carries a larger lower amount of critical dissipation. The first fact leaves room in finite time; the second is a lower cost, not the missing upper budget for D_* .

That unpaid budget cannot remain at a fixed range of frequencies. For $K \geq 1$, set

$$D_{\leq K}(t) := \int_{|\xi| \leq K} |\xi|^3 |\hat{u}(t, \xi)|^2 d\xi.$$

Since $|\xi|^3 \leq K|\xi|^2$ on this region, energy pays

$$\int_0^{T_*} D_{\leq K}(t) dt \leq K \int_0^{T_*} Z(t) dt \leq \frac{K \|u_0\|_2^2}{2\nu} < \infty. \quad (4.1.21)$$

If D_* diverges, its unpaid part must move beyond every fixed K . The missing estimate is an ultraviolet control on the same narrowing history, not a failure of low frequencies or of overlap compatibility.

The size of this first gap is visible from what the first rectangle would do. The adjacent trace lemma turns (4.1.1) into

$$u \in C([0, T_*]; H^1).$$

Since $H^1(\mathbb{R}^3) \hookrightarrow L^3(\mathbb{R}^3)$, it follows that

$$\sup_{0 \leq t \leq T_*} \|u(t)\|_3 < \infty. \quad (4.1.22)$$

But the endpoint theorem of Escauriaza, Seregin, and Šverák gives the opposite conclusion at a finite maximal time (Escauriaza, Seregin, and Šverák 2003):

$$T_* < \infty \implies \sup_{0 < t < T_*} \|u(t)\|_3 = \infty. \quad (4.1.23)$$

So if the first rectangle reaches the endpoint, it already rules out the alleged last time. The larger intersection is not needed to repeat that contradiction; its different work is to carry every finite response and spatial derivative of the same history to one common terminal datum.

To make that larger conclusion, the same cutoff-independent control is needed at every finite coordinate. We state that remaining input at the point where the argument begins to use it.

Assumption 4.1.2 (Whole-terminal rectangle condition). Let $u_0 \in \mathcal{S}_\sigma$, let $\nu > 0$, and let (u, p) be the maximal classical solution generated by (u_0, ν) on $[0, T_*)$. If $T_* < \infty$, then for every finite $N, M \in \mathbb{N}_0$,

$$\sup_{0 < T < T_*} \mathfrak{R}_{N,M}(u; T) < \infty. \quad (4.1.24)$$

The associated pressure rows are generated by (2.2.6) with the fixed decaying normalization.

Here (N, M) is fixed before the supremum over $T < T_*$ is taken. The bound may depend on that pair; no constant uniform over all derivative orders is asserted. Every rectangle belongs to the same history, and larger rectangles return the identical derivatives on their overlaps.

Suppose now that the maximal history ends at a finite T_* . On this alleged branch, Assumption 4.1.2 supplies (4.1.24) on that branch.

Fix N and M . Every term in (3.4.5) is an integral of a nonnegative squared norm. For any one integrand f , the intervals $(0, T)$ exhaust I_* , so

$$\int_{I_*} f(t) dt = \lim_{T \uparrow T_*} \int_0^T f(t) dt.$$

The whole-terminal estimate keeps that limit finite for every control in the chosen rectangle:

$$V_n \in L^2(I_*; H^{M+1}), \quad V_{n+1} = \partial_t V_n \in L^2(I_*; H^{M-1}), \quad 0 \leq n \leq N. \quad (4.1.25)$$

The pressure part of Proposition A.4.1, using the Riesz bounds and finite-order Sobolev products, gives the matching rows

$$Q_n \in L^1(I_*; H^M), \quad 0 \leq n \leq N. \quad (4.1.26)$$

Together these velocity and pressure responses form $\mathcal{R}_{N,M}[u_0]$ on the whole interval I_* .

Now enlarge the rectangle. The fluid does not acquire a second velocity, and a response already reached in the smaller rectangle does not change its value. If $N' \leq N$ and $M' \leq M$,

let $\rho_{N',M'}^{N,M}$ forget the extra rows and read the retained ones at the lower Sobolev height. Every retained entry is still the same derivative of the same u , so

$$\rho_{N',M'}^{N,M} \mathcal{R}_{N,M}[u_0] = \mathcal{R}_{N',M'}[u_0]. \quad (4.1.27)$$

That equality is what lets the intersection be formed: every finite view of the history agrees with every larger view wherever they see the same response. The whole compatible family is

$$\mathcal{I}[u_0] := \left(\mathcal{R}_{N,M}[u_0] \right)_{N,M \in \mathbb{N}_0}, \quad \rho_{N',M'}^{N,M} \mathcal{R}_{N,M}[u_0] = \mathcal{R}_{N',M'}[u_0]. \quad (4.1.28)$$

The notation creates no new field. Given any finite temporal order and spatial height, choose one rectangle large enough to see that derivative; every larger rectangle returns the same derivative of the same u .

Proposition 4.1.3 (The mixed intersection). *On the finite branch just fixed, the velocity component of $\mathcal{I}[u_0]$ satisfies*

$$u \in \bigcap_{r=0}^{\infty} \bigcap_{m=0}^{\infty} H^r(I_*; H^m(\mathbb{R}^3)). \quad (4.1.29)$$

In particular,

$$u \in \bigcap_{m=0}^{\infty} L^2(I_*; H^m(\mathbb{R}^3)). \quad (4.1.30)$$

Proof. Fix finite r, m , and choose $N \geq r$ and $M \geq m$. The upper edge of this one rectangle gives

$$\partial_t^n u \in L^2(I_*; H^{M+1}) \hookrightarrow L^2(I_*; H^m) \quad (0 \leq n \leq r).$$

These are the distributional time derivatives of the same velocity field, hence $u \in H^r(I_*; H^m)$. Because r and m were arbitrary, every finite derivative belongs to the intersection; $r = 0$ gives (4.1.30). $\square$

4.2 The intersection reaches one terminal datum

Keep the alleged finite branch in view. Under Assumption 4.1.2, the mixed intersection does not end as a list of norms. Its adjacent controls force every spatial reading of the same velocity to arrive at the same time slice.

Proposition 4.2.1 (One terminal datum from the mixed intersection). *Suppose $T_* < \infty$ and the maximal history obeys Assumption 4.1.2. Then*

$$u_* := \lim_{t \uparrow T_*} u(t) \in H_\sigma^\infty(\mathbb{R}^3) = \bigcap_{m < \infty} H_\sigma^m(\mathbb{R}^3). \quad (4.2.1)$$

Every temporal response has one trace

$$V_N^* := \lim_{t \uparrow T_*} \partial_t^N u(t) \in H^\infty(\mathbb{R}^3), \quad N \in \mathbb{N}_0,$$

and those traces retain the normalized pressure and velocity recurrences

$$\begin{aligned} Q_N^* &= R_i R_j \sum_{a=0}^N \binom{N}{a} V_{a,i}^* V_{N-a,j}^*, \\ V_{N+1}^* &= \nu \Delta V_N^* - \sum_{a=0}^N \binom{N}{a} (V_a^* \cdot \nabla) V_{N-a}^* - \nabla Q_N^*. \end{aligned}$$

Proof. Proposition 4.1.3 gives

$$u \in \bigcap_{r,m < \infty} H^r(I_*; H^m(\mathbb{R}^3)).$$

Hold one spatial height m fixed. The temporal row $r = 1$ puts u in $H^1(I_*; H^m)$, so the actual history has an H^m limit as $t \uparrow T_*$. A higher spatial row cannot produce a different limit: after embedding it into H^m , both limits come from the same $u(t)$. The complete-row argument in Lemma A.4.2 makes these limits the one datum in (4.2.1).

Now hold N fixed. The row $r = N + 1$ places $V_N = \partial_t^N u$ in $H^1(I_*; H^m)$ for every finite m . Its trace is $V_N^* \in H^\infty$. Passing to the limit in the two recurrences identifies the pressure and the next velocity response exactly as written above; this is the pressure-complete endpoint jet proved in Proposition A.4.3. $\square$

The alleged last time has now become a genuine smooth time slice. Do not restart it yet. First return, on this same finite branch, to the critical height and to the gradient that stretches the vortex. Every conclusion through Section 4.5 still depends on the whole-terminal rectangle condition; none of them derives that condition from (u_0, ν) .

4.3 The critical-height roof

On this alleged finite branch, once the H^1 trace reaches the terminal face, the critical height cannot keep climbing underneath it. Indeed, $H = \frac{1}{2} \|\Lambda^{1/2} u\|_2^2$, so

$$H_{\max} := \sup_{0 \leq t \leq T_*} H(t) \leq \frac{1}{2} \sup_{0 \leq t \leq T_*} \|u(t)\|_{H^1}^2 < \infty. \quad (4.3.1)$$

Start at any positive height H_0 , double it successively by $L_k := 2^k H_0$, and count the levels the same history actually reaches:

$$N_{\text{levels}} := \#\{k \in \mathbb{N}_0 : \text{there is a } t \in [0, T_*] \text{ with } H(t) = L_k\}.$$

The roof leaves room for only finitely many such doublings:

$$N_{\text{levels}} \leq \begin{cases} 0, & H_{\max} < H_0, \\ 1 + \lfloor \log_2(H_{\max}/H_0) \rfloor, & H_{\max} \geq H_0. \end{cases} \quad (4.3.2)$$

This conclusion never asks how quickly one level is crossed. It rules out an infinite succession of critical-height doublings. It does not rule out an infinite succession of smaller widths at the same height: the rescaled profile in Section 2 was built to show exactly that possibility. The roof closes escalation in H ; the geometric picture still has to close concentration at fixed H .

The next rectangle, $\mathcal{R}_{0,2}$, follows the motion below the roof rather than only its maximum:

$$u \in L^2(I_*; H^3), \quad u_t \in L^2(I_*; H^1), \quad u \in C([0, T_*]; H^2). \quad (4.3.3)$$

The two adjacent controls now contain both factors in the derivative

$$H'(t) = \operatorname{Re} \left\langle \Lambda^{1/2} u(t), \Lambda^{1/2} u_t(t) \right\rangle_{L^2},$$

and Cauchy–Schwarz bounds its total variation:

$$\int_0^{T_*} |H'(t)| dt \leq \left\| \Lambda^{1/2} u \right\|_{L_t^2 L_x^2} \left\| \Lambda^{1/2} u_t \right\|_{L_t^2 L_x^2} < \infty. \quad (4.3.4)$$

For a vortex, the decisive entry in $\mathcal{R}_{0,2}$ is $u \in L^2(I_*; H^3)$. In three dimensions $H^3 \hookrightarrow W^{1,\infty}$, whence

$$\int_0^{T_*} \|\nabla u(t)\|_\infty dt \leq T_*^{1/2} \|u\|_{L^2(I_*; H^3)} < \infty.$$

Both vorticity and strain are parts of this gradient. Their accumulated action before T_* is finite in the precise sense

$$\int_0^{T_*} (\|\nabla u(t)\|_\infty + \|\omega(t)\|_\infty + \|S(t)\|_\infty) dt < \infty. \quad (4.3.5)$$

4.4 The two vortex readings

What does a narrowing core mean for the fluid inside it? There are two answers to separate. The first follows the same material particles. The second allows the visible concentration to replace its particles as it moves. Neither exhausts singular geometry, but each makes a different claim that the rectangle bounds can test.

Follow the fixed labels first. Let $X(a, t)$ solve

$$\partial_t X(a, t) = u(X(a, t), t), \quad X(a, 0) = a.$$

Two such trajectories satisfy

$$\frac{d}{dt} |X(a, t) - X(b, t)| \geq -\|\nabla u(t)\|_\infty |X(a, t) - X(b, t)|,$$

and the finite Lipschitz action integrates this differential inequality to

$$|X(a, t) - X(b, t)| \geq |a - b| \exp\left(-\int_0^t \|\nabla u(s)\|_\infty ds\right) > 0 \quad (4.4.1)$$

Because the same row also puts u in $L^1(I_*; L^\infty)$, every trajectory has a limit at T_* . Letting $t \uparrow T_*$ preserves the lower bound:

$$|X(a, T_*) - X(b, T_*)| \geq |a - b| \exp\left(-\int_0^{T_*} \|\nabla u(s)\|_\infty ds\right) > 0.$$

Distinct labels remain spatially distinct at the terminal time. If a drawn core collapses to one point, it cannot still contain exactly the material core with which it began.

Even perfect centring cannot rescue a nonzero circulation from this fixed-label picture. The adjacent trace of $\mathcal{R}_{0,3}$ gives $u \in C([0, T_*]; H^3)$; since $H^3(\mathbb{R}^3) \hookrightarrow W^{1,\infty}(\mathbb{R}^3)$,

$$L_* := \sup_{0 \leq t \leq T_*} \|\nabla u(t)\|_\infty < \infty. \quad (4.4.2)$$

Relative velocity at any chosen centre $x_c(t)$ obeys

$$|u(t, x) - u(t, x_c)| \leq L_* |x - x_c|.$$

Around a circular cross-section $C_r(t)$, the constant centre velocity integrates to zero. What remains is bounded by its Lipschitz variation:

$$\left| \oint_{C_r(t)} u(t, x) \cdot d\ell \right| \leq 2\pi L_* r^2 \longrightarrow 0 \quad (r \downarrow 0). \quad (4.4.3)$$

As r shrinks, the tangential velocity relative to the centre is $O(r)$ and the circuit has length $O(r)$. Their product is $O(r^2)$, so this centred material core cannot carry a nonzero limiting circulation.

Now let membership change. Particle separation no longer tracks the visible core. One precise spectral version of repeated recruitment says that a fixed amount of critical content survives at smaller and smaller wavelengths. That version can be tested in the high-frequency tail of the whole velocity field. For $s > 1/2$ and $K \geq 1$, define

$$H_{>K}(t) := \frac{1}{2} \int_{|\xi| > K} |\xi| |\hat{u}(t, \xi)|^2 d\xi.$$

For every $s > 1/2$, the tail satisfies

$$H_{>K}(t) \leq \frac{1}{2} K^{1-2s} \|u(t)\|_{\dot{H}^s}^2. \quad (4.4.4)$$

Choose one integer $m \geq s$. The compatible assembly and adjacent trace proved in Proposition 4.1.3 and theorem A.3.1, followed by $H^m \hookrightarrow \dot{H}^s$, make the estimate uniform in time and force

$$\lim_{K \rightarrow \infty} \sup_{0 \leq t \leq T_*} H_{>K}(t) = 0. \quad (4.4.5)$$

Activity may move between scales, but it cannot leave a fixed amount of critical height beyond every cutoff. This rules out the spectral version just stated. It does not prove that every geometrically narrowing core with changing membership must create such a tail; that geometric-to-Fourier implication would require a separate localization argument. What the two calculations establish is narrower: fixed labels cannot collapse and their centred circulation vanishes, while a changing-membership model cannot preserve a fixed critical tail arbitrarily far into the ultraviolet.

4.5 The singularity conditions fail at both scales

The local signature disappears at the same alleged endpoint. The common terminal trace in Lemma A.4.2 puts u in $C([0, T_*]; H^m)$ at every finite height. Choose $m > 3/2$. Since $H^m(\mathbb{R}^3)$ is an algebra, the normalized pressure formula $p = R_i R_j (u_i u_j)$ gives the common ceiling

$$K_* := \sup_{0 \leq t \leq T_*} (\|u(t)\|_{L^\infty} + \|p(t)\|_{L^\infty}) < \infty. \quad (4.5.1)$$

Now choose $z_* = (x_*, T_*)$ on the terminal face. For $\rho > 0$ small enough, place the backward parabolic cylinder

$$Q_\rho(z_*) = B_\rho(x_*) \times (T_* - \rho^2, T_*) \subset \mathbb{R}^3 \times (0, T_*].$$

Its spatial volume and time length contribute ρ^5 . Dividing by the critical factor ρ^2 still leaves three powers of the radius:

$$\rho^{-2} \iint_{Q_\rho(z_*)} (|u|^3 + |p|^{3/2}) \, dx \, dt \leq C \rho^3 (K_*^3 + K_*^{3/2}) \longrightarrow 0 \quad (\rho \downarrow 0). \quad (4.5.2)$$

A Caffarelli–Kohn–Nirenberg singular point would require a positive scale-invariant amount of velocity-pressure content to survive along shrinking cylinders (Caffarelli, Kohn, and Nirenberg 1982). Here that content vanishes.

This is the return promised by the vortex picture. The alleged endpoint has become a smooth time slice. The critical height cannot climb through infinitely many doublings, but that roof is not being used as a bound on the number of spatial scales. The remaining calculations answer the precise fixed-height scenarios posed above: the stretching action is integrable, a fixed material core cannot collapse its labels, a fixed critical tail cannot survive arbitrarily far into high frequency, and the local velocity-pressure content loses the mass a singular point would need at every smaller scale.

4.6 Restarting the same history

The terminal slice still belongs to the history that began at u_0 . It can now be used as initial data without changing the velocity, the pressure normalization, or any response by which the history arrived there.

Theorem 4.6.1 (Conditional global closure from whole-terminal rectangles). *Let $\nu > 0$ and $u_0 \in \mathcal{S}_\sigma$, and assume that the maximal classical history generated by (u_0, ν) obeys Assumption 4.1.2. Then*

$$T_* = \infty, \quad u, p \in C^\infty(\mathbb{R}^3 \times [0, \infty)). \quad (4.6.1)$$

For every $t \geq 0$,

$$\frac{1}{2} \|u(t)\|_2^2 + \nu \int_0^t \|\nabla u(s)\|_2^2 \, ds = \frac{1}{2} \|u_0\|_2^2. \quad (4.6.2)$$

Proof. Suppose $T_* < \infty$. The rectangle condition activates on this branch, and Proposition 4.2.1 gives the common smooth datum u_* and its pressure-complete left jet. Use u_* as initial data at time T_* . The smooth local construction in Lemma A.4.4 gives a right-hand history on $[T_*, T_* + \delta]$. Its recurrence begins from the same datum and the same pressure normalization, so its jet agrees with the left-hand traces. The join is smooth. Strong uniqueness makes it the continuation of the history generated by u_0 , contradicting the definition of T_* .

Hence $T_* = \infty$, and the normalized pair is smooth on every finite time interval. The kinetic-energy identity in Proposition A.2.1 holds on every initial segment and gives (4.6.2). In particular,

$$\sup_{t \geq 0} \int_{\mathbb{R}^3} |u(x, t)|^2 dx \leq \int_{\mathbb{R}^3} |u_0(x)|^2 dx < \infty.$$

This is the global history required in (1.4). □

The restart closes the implication from whole-terminal rectangles to global smoothness. The unpaid direction is still the one isolated in (4.1.10): this manuscript has not derived even the lowest rectangle, or equivalently $D_* < \infty$, from an arbitrary pair (u_0, ν) .

A Technical Foundations

The body follows the argument in the order the obstruction appears. The supporting proofs collected here launch and restart the normalized history, place a strong trace between two adjacent spacetime rows, and carry the compatible terminal rows to one endpoint jet.

A.1 The local construction behind the maximal history

One local theorem must do two jobs without changing its clock. It constructs the initial history and later restarts it from the terminal datum. The clock is set by ν and the datum's H^3 size; any higher regularity already carried by that datum survives on the same interval.

For real s , let

$$H_\sigma^s(\mathbb{R}^3) := \{v \in H^s(\mathbb{R}^3; \mathbb{R}^3) : \nabla \cdot v = 0\}.$$

Proposition A.1.1 (Local construction and continuation alternative). *Fix $t_0 \geq 0$, $\nu > 0$, and $a \in H_\sigma^3(\mathbb{R}^3)$. A time $\delta = \delta(\nu, \|a\|_{H^3}) > 0$ exists for which there is a unique normalized-pressure pair (w, q) satisfying*

$$w \in C([t_0, t_0 + \delta]; H_\sigma^3) \cap L^2(t_0, t_0 + \delta; H^4), \quad w_t \in L^2(t_0, t_0 + \delta; H^2).$$

If the datum also lies in H_σ^m for an integer $m > 3$, this same interval carries

$$w \in C([t_0, t_0 + \delta]; H_\sigma^m) \cap L^2(t_0, t_0 + \delta; H^{m+1}), \quad w_t \in L^2(t_0, t_0 + \delta; H^{m-1}).$$

These local pairs agree on overlaps and form one maximal history. If its maximal time is finite, its H^3 norm is unbounded along times approaching that endpoint.

Proof of Proposition A.1.1. Keep t_0, ν, a fixed as above and measure all real Sobolev orders with

$$\|f\|_{H^s}^2 := \int_{\mathbb{R}^3} \langle \xi \rangle^{2s} |\widehat{f}(\xi)|^2 d\xi, \quad \langle \xi \rangle = (1 + |\xi|^2)^{1/2}.$$

Write $\mathbb{P}$ for the Leray projection and R_j for the Riesz transform with symbol $i\xi_j/|\xi|$. Each acts boundedly on every H^s .

The three rows forced by the nonlinearity and the heat equation

The construction must first absorb one derivative of $w \otimes w$. In three dimensions,

$$H^3(\mathbb{R}^3) \hookrightarrow W^{1,\infty}(\mathbb{R}^3), \quad \|fg\|_{H^3} \leq C \|f\|_{H^3} \|g\|_{H^3},$$

so an H^3 velocity is Lipschitz and products stay in H^3 . After the divergence and Leray projection this gives

$$\|\mathbb{P}\nabla \cdot (f \otimes g)\|_{H^2} \leq C \|f\|_{H^3} \|g\|_{H^3}. \quad (\text{A.1.1})$$

Leibniz' rule proves the estimate for integer derivatives, using $H^2 \hookrightarrow L^\infty$ and $H^1 \hookrightarrow L^6$; density then removes the smoothness assumption.

Once the force lies in H^2 , the heat equation places one parabolic derivative above the base state and the time derivative one below it. Thus $z \in L_t^2 H_x^4$ and $z_t \in L_t^2 H_x^2$, with H^3 exactly between them. For any such v ,

$$v \in L^2(J; H^4), \quad v_t \in L^2(J; H^2),$$

frequency truncation followed by the fundamental theorem of calculus and Cauchy–Schwarz yields

$$\left| \|v(t)\|_{H^3}^2 - \|v(r)\|_{H^3}^2 \right| \leq 2 \|v_t\|_{L^2(r,t;H^2)} \|v\|_{L^2(r,t;H^4)}. \quad (\text{A.1.2})$$

The right-hand side vanishes with the length of the time interval. Removing the frequency cutoff gives a unique representative in $C(J; H^3)$, so the three rows of the strong class are

$$C_t H_x^3 \cap L_t^2 H_x^4, \quad \partial_t v \in L_t^2 H_x^2.$$

The continuous middle trace is what lets the equation remember its initial value while the heat operator gains one square-integrable derivative.

A linear estimate aligned with those rows

On $J_\delta = [t_0, t_0 + \delta]$, $0 < \delta \leq 1$, consider

$$z_t - \nu \Delta z = F, \quad z(t_0) = a, \quad F \in L^2(J_\delta; H^2).$$

At the H^3 level the force and solution occupy the adjacent spaces H^2 and H^4 . Their Fourier pairing is

$$\langle F, z \rangle_{H^2, H^4; H^3} := \int_{\mathbb{R}^3} \langle \xi \rangle^6 \widehat{F}(\xi) \cdot \overline{\widehat{z}(\xi)} d\xi.$$

The weight factors as $\langle \xi \rangle^6 = \langle \xi \rangle^2 \langle \xi \rangle^4$ and Cauchy–Schwarz gives

$$|\langle F, z \rangle_{H^2, H^4; H^3}| \leq \|F\|_{H^2} \|z\|_{H^4}. \quad (\text{A.1.3})$$

For a smooth solution the corresponding energy row is

$$\frac{1}{2} \frac{d}{dt} \|z\|_{H^3}^2 + \nu \|\nabla z\|_{H^3}^2 = \text{Re} \langle F, z \rangle_{H^2, H^4; H^3}.$$

The inhomogeneous norm separates the dissipative part exactly:

$$\|z\|_{H^4}^2 = \|z\|_{H^3}^2 + \|\nabla z\|_{H^3}^2.$$

Insert (A.1.3), absorb by Young's inequality, and apply Gronwall to obtain

$$\|z\|_{C(J_\delta; H^3)} + \|z\|_{L^2(J_\delta; H^4)} \leq C_\nu \left(\|a\|_{H^3} + \|F\|_{L^2(J_\delta; H^2)} \right).$$

The equation then controls the missing time row as well:

$$\|z\|_{C(J_\delta; H^3)} + \|z\|_{L^2(J_\delta; H^4)} + \|z_t\|_{L^2(J_\delta; H^2)} \leq C_\nu \left(\|a\|_{H^3} + \|F\|_{L^2(J_\delta; H^2)} \right). \quad (\text{A.1.4})$$

No constant here deteriorates as δ shrinks within $(0, 1]$. Frequency-truncate a and F , solve the smooth equations, and apply (A.1.4) to differences. The Cauchy limit solves the untruncated problem, while (A.1.2) recovers its prescribed strong H^3 value at t_0 . Applying the estimate to the difference of two solutions proves uniqueness.

Closing the nonlinear map

Set

$$X_\delta := C(J_\delta; H_\sigma^3) \cap L^2(J_\delta; H^4), \quad \|w\|_{X_\delta} := \|w\|_{CH^3} + \|w\|_{L^2H^4}.$$

The projected equation defines the map

$$\Phi(w)(t) := e^{\nu(t-t_0)\Delta}a - \int_{t_0}^t e^{\nu(t-s)\Delta}\mathbb{P}\nabla \cdot (w \otimes w)(s) ds. \quad (\text{A.1.5})$$

All abbreviated time norms below refer to J_δ . The product estimate and the factor $\delta^{1/2}$ from the interval give

$$\|\mathbb{P}\nabla \cdot (w \otimes w)\|_{L^2H^2} \leq C\delta^{1/2} \|w\|_{CH^3}^2, \quad (\text{A.1.6})$$

$$\|\mathbb{P}\nabla \cdot (w \otimes w - v \otimes v)\|_{L^2H^2} \leq C\delta^{1/2} (\|w\|_{CH^3} + \|v\|_{CH^3}) \|w - v\|_{CH^3}. \quad (\text{A.1.7})$$

The linear estimate turns these force bounds into

$$\|\Phi(w)\|_{X_\delta} \leq C_\nu \|a\|_{H^3} + C_\nu C\delta^{1/2} \|w\|_{X_\delta}^2$$

and

$$\|\Phi(w) - \Phi(v)\|_{X_\delta} \leq C_\nu C\delta^{1/2} (\|w\|_{X_\delta} + \|v\|_{X_\delta}) \|w - v\|_{X_\delta}.$$

When $a \neq 0$, set $R = 2C_\nu \|a\|_{H^3}$ and choose $\delta \leq 1$ with $4C_\nu C\delta^{1/2}R \leq 1$. Then Φ maps the closed radius- R ball into itself and contracts it by at least a factor of two. For $a = 0$, the fixed point is identically zero. Banach's theorem gives $w \in X_\delta$, with a choice satisfying

$$\delta \geq \min \left\{ 1, c_\nu (1 + \|a\|_{H^3})^{-2} \right\}. \quad (\text{A.1.8})$$

The constant $c_\nu > 0$ depends only on the fixed viscosity and the chosen Sobolev normalization.

Uniqueness is not confined to that ball. If $w, v \in X_\delta$ share the datum, apply the linear and difference estimates to $w - v$ on an initial subinterval short enough to absorb the coefficient. The solutions agree there. Their common endpoint starts the same argument on the next subinterval, and a finite partition covers J_δ . The heat semigroup and $\mathbb{P}$ preserve divergence-free fields; the projected equation also supplies

$$w_t \in L^2(J_\delta; H^2). \quad (\text{A.1.9})$$

Restoring the pressure to the same solution

For the constructed velocity, set

$$q := R_i R_j (w_i w_j). \quad (\text{A.1.10})$$

The algebra property and Riesz bounds imply

$$q \in C(J_\delta; H^3).$$

For $F = \nabla \cdot (w \otimes w)$, the Fourier symbols identify $\mathbb{P}F = F + \nabla q$. Hence the projected fixed point is the pressure-complete solution

$$w_t + (w \cdot \nabla)w + \nabla q = \nu \Delta w, \quad \nabla \cdot w = 0, \quad w(t_0) = a.$$

Membership of $q(t)$ in $H^3(\mathbb{R}^3)$ removes the whole-space additive constant. Two normalized pressures for the same velocity differ by an L^2 function with zero gradient and hence coincide. The uniqueness already proved is consequently uniqueness of the pair (w, q) .

Higher regularity without a shorter clock

Fix an integer $m \geq 3$ and temporarily take a smooth. Differentiate the projected equation by every ∂^α , $|\alpha| \leq m$, pair with the same derivative of w , and sum. Divergence-free transport cancels at the top order:

$$\left((w \cdot \nabla) \partial^\alpha w, \partial^\alpha w \right)_2 = 0.$$

Every remaining Leibniz term differentiates the transporting factor at least once. The embeddings $H^2 \hookrightarrow L^\infty$ and $H^1 \hookrightarrow L^6$ then give

$$\left(\sum_{|\alpha| \leq m} \|[\partial^\alpha, w \cdot \nabla]w\|_2^2 \right)^{1/2} \leq C_m \|\nabla w\|_\infty \|w\|_{H^m}.$$

Thus the differentiated energy obeys

$$\frac{1}{2} \frac{d}{dt} \|w\|_{H^m}^2 + \nu \|\nabla w\|_{H^m}^2 \leq C_m \|\nabla w\|_\infty \|w\|_{H^m}^2. \quad (\text{A.1.11})$$

The already constructed $C_t H_x^3$ solution has integrable Lipschitz norm on this same interval. Gronwall yields

$$\|w(t)\|_{H^m}^2 \leq \|a\|_{H^m}^2 \exp \left(2C_m \int_{t_0}^t \|\nabla w(s)\|_\infty ds \right). \quad (\text{A.1.12})$$

Retaining the dissipative term gives $w \in L^2(J_\delta; H^{m+1})$. At order $m-1$, the nonlinear term lies in $L^2(J_\delta; H^{m-1})$, so the equation also gives

$$w_t \in L^2(J_\delta; H^{m-1}).$$

The adjacent trace at height m completes the three rows:

$$w \in C(J_\delta; H^m) \cap L^2(J_\delta; H^{m+1}), \quad w_t \in L^2(J_\delta; H^{m-1}). \quad (\text{A.1.13})$$

The pressure formula and the H^m algebra property put q in $C(J_\delta; H^m)$ as well.

For general $a \in H_\sigma^m$, approximate it by divergence-free frequency cutoffs a_k . Their H^3 norms share a bound controlled by $\|a\|_{H^3}$; hence (A.1.8) puts every smooth approximation on one common interval. The fixed-point estimate identifies the H^3 limit with w , while the persistence inequalities bound all three higher rows uniformly. Weak compactness then identifies their limits with the same w . The adjacent trace gives continuity in H^m ; the energy

inequality at t_0 , together with weak convergence to a , upgrades the initial convergence to strong convergence in H^m .

The interval never depends on m ; it was chosen once from ν and $\|a\|_{H^3}$. If $a \in H^\infty_\sigma$, repeat the argument at each finite height. Uniqueness in H^3 keeps every height on the same pair, and the pressure formula and momentum equation then recover every finite time derivative. Spatial embedding makes (w, q) smooth throughout the local space–time slab.

Joining the local pieces and locating the only obstruction

Begin at t_0 and extend through every interval allowed by the construction. Strong uniqueness makes the velocity and normalized pressure identical on overlaps, producing one pressure-complete history on a maximal interval $[t_0, T_{\max})$.

Suppose $T_{\max} < \infty$ and

$$\sup_{t_0 \leq t < T_{\max}} \|w(t)\|_{H^3} \leq M.$$

If its displayed H^3 bound were finite, the lifespan estimate would supply the same positive amount of future time

$$\delta_M := \min \{1, c_\nu(1 + M)^{-2}\}$$

from every later datum $w(t)$. Choose $t > T_{\max} - \delta_M/2$. Restarting at $w(t)$ then crosses $T_{\max}$, and uniqueness glues that restart to the supposed maximal history. The contradiction proves

$$T_{\max} < \infty \implies \limsup_{t \uparrow T_{\max}} \|w(t)\|_{H^3} = \infty. \quad (\text{A.1.14})$$

Every higher order already present in the datum persists along each piece, because every piece uses this same H^3 -controlled clock. In particular, $a \in \mathcal{S}_\sigma \subset H^\infty_\sigma$ generates the smooth normalized maximal history claimed above. $\square$

A.2 The kinetic-energy identity

Proposition A.2.1 (Classical kinetic-energy identity). *Let (u, p) be a sufficiently decaying smooth solution of (1.3). At every time in its interval of existence,*

$$\|u(t)\|_{L^2}^2 + 2\nu \int_0^t \|\nabla u(s)\|_{L^2}^2 ds = \|u_0\|_{L^2}^2. \quad (\text{A.2.1})$$

Proof. Pair the momentum equation with u in space. Incompressibility removes the net transport and pressure contributions:

$$\int_{\mathbb{R}^3} (u \cdot \nabla) u \cdot u \, dx = \frac{1}{2} \int_{\mathbb{R}^3} u \cdot \nabla |u|^2 \, dx = 0, \quad \int_{\mathbb{R}^3} \nabla p \cdot u \, dx = - \int_{\mathbb{R}^3} p \nabla \cdot u \, dx = 0.$$

Integration by parts leaves viscosity with the sign

$$\nu \int_{\mathbb{R}^3} \Delta u \cdot u \, dx = -\nu \|\nabla u\|_{L^2}^2.$$

Integrating the resulting identity in time gives (A.2.1). $\square$

A.3 The trace between adjacent rows

Lemma A.3.1 (Adjacent Hilbert-scale trace). *Let $s \in \mathbb{R}$ and suppose*

$$v \in L^2(I_*; H^{s+1}), \quad v_t \in L^2(I_*; H^{s-1}).$$

Then v has one representative in $C([0, T_]; H^s)$, and for $0 \leq a \leq b \leq T_*$,*

$$\left| \|v(b)\|_{H^s}^2 - \|v(a)\|_{H^s}^2 \right| \leq 2 \|v_t\|_{L^2(a,b; H^{s-1})} \|v\|_{L^2(a,b; H^{s+1})}. \quad (\text{A.3.1})$$

Proof. Let $A = (1 - \Delta)^{1/2}$. Time mollification and frequency truncation make the identity

$$\|v(b)\|_{H^s}^2 - \|v(a)\|_{H^s}^2 = 2 \operatorname{Re} \int_a^b \left\langle A^{s-1} v_t(t), A^{s+1} v(t) \right\rangle_{L^2} dt \quad (\text{A.3.2})$$

legitimate for smooth approximants. Cauchy–Schwarz gives (A.3.1), and the two assumed L^2 rows pass the right-hand side to the limit. The limit identity makes the H^s norm continuous. The same approximation gives weak H^s continuity, so the representative is strongly continuous. Any two continuous representatives agree everywhere because they agree almost everywhere. $\square$

A.4 From terminal rows to one restart datum

Proposition A.4.1 (Terminal realization of the rectangle estimate). *Let $0 < T_* < \infty$, and let (u, p) be the maximal normalized history on $I_* = (0, T_*)$ generated by (u_0, ν) . Assume Assumption 4.1.2. For every finite $N, M \in \mathbb{N}_0$,*

$$\sum_{n=0}^N \|V_n\|_{L^2(I_*; H^{M+1})}^2 + \sum_{n=0}^N \|V_{n+1}\|_{L^2(I_*; H^{M-1})}^2 < \infty, \quad (\text{A.4.1})$$

$$Q_n \in L^1(I_*; H^M), \quad 0 \leq n \leq N.$$

All shared entries are the same distributional derivatives of u , and the pressure rows retain the normalization (2.2.5).

Proof. Fix (N, M) . For each nonnegative integrand in (3.4.5), monotone convergence and the cutoff-uniform bound give

$$\int_{I_*} f(t) dt = \lim_{T \uparrow T_*} \int_0^T f(t) dt < \infty.$$

The finite sum yields both velocity rows in (A.4.1). The pressure recurrence (2.2.6), boundedness of the Riesz transforms, Sobolev multiplication, and Cauchy–Schwarz in time give

$$\|Q_n\|_{L^1(I_*; H^M)} \leq C_{M,n} \sum_{a=0}^n \|V_a\|_{L^2(I_*; H^{M+1})} \|V_{n-a}\|_{L^2(I_*; H^{M+1})} < \infty.$$

Changing the rectangle drops rows or reads them in a lower Sobolev space; it does not change a shared derivative or normalized pressure. $\square$

Lemma A.4.2 (The complete spatial row reaches every endpoint topology). *Let $0 < T_* < \infty$, and let (u, p) be the fixed normalized history on $I_* = (0, T_*)$. If*

$$u \in \bigcap_{m=0}^{\infty} L^2(I_*; H^m(\mathbb{R}^3)),$$

then for every $m \in \mathbb{N}_0$,

$$u_t \in L^1(I_*; H^m), \quad u \in W^{1,1}(I_*; H^m) \hookrightarrow C([0, T_*]; H^m). \quad (\text{A.4.2})$$

All of these traces meet at one

$$u_* := u(T_*) \in H_{\sigma}^{\infty}(\mathbb{R}^3).$$

Proof. The pressure-complete transport response is

$$(u \cdot \nabla)u + \nabla p = \mathbb{P} \nabla \cdot (u \otimes u).$$

Sobolev multiplication and boundedness of the Leray and Riesz multipliers give

$$\|\mathbb{P} \nabla \cdot (u \otimes u)\|_{H^m} \leq C_m \|u\|_{L^2(I_*; H^{m+2})}^2.$$

Because $T_* < \infty$, the complete row yields

$$\begin{aligned} \|\mathbb{P} \nabla \cdot (u \otimes u)\|_{L^1(I_*; H^m)} &\leq C_m \|u\|_{L^2(I_*; H^{m+2})}^2 < \infty, \\ \|\Delta u\|_{L^1(I_*; H^m)} &\leq T_*^{1/2} \|u\|_{L^2(I_*; H^{m+2})} < \infty. \end{aligned}$$

The momentum equation puts u_t in $L^1(I_*; H^m)$. For $0 \leq s < t < T_*$,

$$\|u(t) - u(s)\|_{H^m} \leq \nu(t-s)^{1/2} \|u\|_{L^2((s,t); H^{m+2})} + C_m \|u\|_{L^2((s,t); H^{m+2})}^2.$$

Both terms vanish as $s, t \uparrow T_*$, so the same $u(t)$ is Cauchy in every H^m . The Bochner fundamental theorem of calculus gives the continuous representatives. If $k > m$, the H^k trace embedded into H^m and the native H^m trace agree almost everywhere and hence everywhere by continuity. Their terminal values are one element of $\bigcap_m H^m$; divergence and finite energy pass to the limit. $\square$

Assume Assumption 4.1.2. Applying the adjacent trace to every temporal coordinate gives

$$V_N^* := \lim_{t \uparrow T_*} \partial_t^N u(t) \in H^\infty(\mathbb{R}^3), \quad N \in \mathbb{N}_0. \quad (\text{A.4.3})$$

Proposition A.4.3 (Pressure-complete endpoint jet). *Assume the hypotheses of Proposition A.4.1, and let u_* and $(V_N^*)_{N \geq 0}$ be the common traces furnished by Lemma A.4.2 and (A.4.3). Then u_* determines one sequence $(V_N^*, Q_N^*)_{N \geq 0} \subset H^\infty \times H^\infty$ by*

$$\begin{aligned} V_0^* &= u_*, \\ Q_N^* &= R_i R_j \sum_{a=0}^N \binom{N}{a} V_{a,i}^* V_{N-a,j}^*, \end{aligned} \quad (\text{A.4.4})$$

$$V_{N+1}^* = \nu \Delta V_N^* - \sum_{a=0}^N \binom{N}{a} (V_a^* \cdot \nabla) V_{N-a}^* - \nabla Q_N^*. \quad (\text{A.4.5})$$

This sequence is the family of left traces of the original normalized velocity and pressure history.

Proof. The space $H^\infty = \bigcap_m H^m$ is stable under multiplication and spatial differentiation, and Riesz transforms preserve each Sobolev coordinate. The two recurrences determine a unique next pair at every finite order. Fix a finite target height and take the trace convergence high enough for each product and derivative. Passing to the limit in (2.2.6) and (2.2.4), then inducting in N , identifies the generated sequence with the left traces. The decaying Riesz formula also preserves the pressure normalization at the terminal face. $\square$

Lemma A.4.4 (Smooth local restart and uniqueness). *Let $t_0 \geq 0$ and $a \in H_\sigma^\infty(\mathbb{R}^3)$. There are $\delta > 0$ and a unique smooth normalized pair (w, q) on $[t_0, t_0 + \delta] \times \mathbb{R}^3$, with $w(t_0) = a$ and*

$$q = R_i R_j (w_i w_j).$$

All finite spatial Sobolev orders persist on this same interval, and the temporal jet at t_0 is generated by (A.4.4)–(A.4.5).

Proof. Translate time and apply Proposition A.1.1 to a . Its lifespan depends only on ν and $\|a\|_{H^3}$, while persistence carries every finite height of a on that interval. Uniqueness fixes the normalized pair. Differentiating at the initial face and using the Riesz pressure formula gives the endpoint recurrences. $\square$

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